A Relative Two-Evaluation Criterion for Countable Capture

Run `fa_banach_001`, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative $C(K)$ criterion and the von Neumann algebra criterion, it asks for a noncommutative C^*-algebra version:

> $\{U_\alpha\}_{\alpha \in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Definition 1. *Let A be a unital C^*-algebra. We say that A has the* relative two-evaluation property *if for every separable unital C^*-subalgebra $B \subseteq A$, there are a self-adjoint element $u \in A$ with $\|u\| = 1$, a unital C^*-algebra D, and contractive linear maps*

$$\theta_+, \theta_- : C^*(B, u) \to D$$

such that the restrictions $\theta_\pm|_B$ are isometric and agree, while

$$\theta_+(u) = 1_D, \qquad \theta_-(u) = -1_D.$$

Theorem 1. *If a unital C^*-algebra A has the relative two-evaluation property, then A has countable capture. Consequently A fails BCP.*

Corollary 1 (tensorial sufficient condition)**.** *Suppose that for every separable unital C^*-subalgebra $B \subseteq A$ there is a self-adjoint $u \in A$ with $\|u\| = 1$, $-1, 1 \in \sigma(u)$, $u \in B' \cap A$, and the canonical multiplication map*

$$B \otimes_{\min} C^*(u) \to C^*(B, u)$$

is an isomorphism. Then A has countable capture and fails BCP.

Idea of the Proof

The countable-capture characterization says that it is enough, after seeing a countable centre set C, to find $u \in S_A$ such that

$$\operatorname{dist}(c, \mathbb{C}u) = \|c\| \qquad (c \in C).$$

Put the countable set inside a separable subalgebra B. The two evaluation maps turn $c - \lambda u$ into two elements

$$\theta(c) - \lambda 1, \qquad \theta(c) + \lambda 1.$$

The maximum of their norms is at least $\|\theta(c)\| = \|c\|$, because their sum is $2\theta(c)$. This forces

$$\|c - \lambda u\| \geq \|c\|$$

for every scalar λ, hence gives countable capture.

Proof

Proof of Theorem 1. Let $C \subset A$ be countable. Let $B = C^*(1, C)$, a separable unital C^*-subalgebra of A. Choose u, D, θ_+, θ_- from the relative two-evaluation property. Since $\theta_\pm|_B$ agree, write $\theta : B \to D$ for their common restriction. This restriction is isometric.

Fix $c \in C$ and $\lambda \in \mathbb{C}$. Contractivity gives

$$\begin{aligned} \|c - \lambda u\| &\geq \max\{\|\theta_+(c - \lambda u)\|, \|\theta_-(c - \lambda u)\|\} \\ &= \max\{\|\theta(c) - \lambda 1\|, \|\theta(c) + \lambda 1\|\}. \end{aligned}$$

By the triangle inequality,

$$2\|c\| = 2\|\theta(c)\| = \|(\theta(c) - \lambda 1) + (\theta(c) + \lambda 1)\| \leq 2 \max\{\|\theta(c) - \lambda 1\|, \|\theta(c) + \lambda 1\|\}.$$

Thus $\|c - \lambda u\| \geq \|c\|$ for every λ. Taking the infimum over λ gives $\operatorname{dist}(c, \mathbb{C}u) \geq \|c\|$, and equality follows from $\lambda = 0$. Therefore A has countable capture by the previous countable-capture characterization. The countable-capture obstruction then implies that A fails BCP. □

Proof of Corollary 1. Let $B \subseteq A$ be separable and choose u as in the hypothesis. Because $-1, 1 \in \sigma(u)$, the evaluation maps

$$\mathrm{ev}_{\pm 1} : C^*(u) \cong C(\sigma(u)) \to \mathbb{C}$$

are characters. Under the isomorphism

$$C^*(B, u) \cong B \otimes_{\min} C^*(u),$$

the maps

$$\theta_\pm = \mathrm{id}_B \otimes \mathrm{ev}_{\pm 1}$$

are contractive and fix B, while sending u to ± 1. Hence the relative two-evaluation property holds, and Theorem 1 applies. □

Scope

This criterion recovers the fresh-coordinate argument for uncountable tensor products. More importantly, it isolates the exact kind of lemma that would settle the remaining arbitrary simple nonseparable C^*-algebra wall: one would need to show that every separable subalgebra admits such a fresh two-evaluation direction, or else construct a counterexample where no such direction exists.

Novelty and Search Bounds

The local run indexes and web searches were checked for countable capture, two-evaluation, infinite tensor products, relative commutants, and BCP for C^*-algebras. No exact statement of this relative criterion was found. The theorem is a distilled abstraction of the uncountable tensor-product capture packet.

Human Review Focus

Reviewers should check:

1. the definition of the two-evaluation property;
2. the distance inequality in Theorem 1;
3. the tensorial sufficient condition in Corollary 1.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Run `fa_banach_001`, lane 19, *When the Countable-Capture BCP Obstruction Exists*, solution packet `2311.14257_countable_capture_characterization`, 2026.

[3] Run `fa_banach_001`, lane 19, *Uncountable Tensor Products and Countable Capture*, solution packet `2311.14257_uncountable_tensor_product_capture`, 2026.